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1. If $y = f(x)$ is a function satisfying the equation $x^3y^2 - 2x + y^3 = 36$, find the tangent of the equation.
2. Let $s = t^3 - 9t^2 + 24t$ describe the position s at time t of an object moving on a straight line. find the velocity & acceleration of object.
3. Find the first and second order derivative of $sec^2 3x$.
4. Let $y = e^x x^3$. find the nth derivative of the function.
5. If $y = \frac{ax+b}{cx+d}$, prove that $2y_1y_3 = 3y_2^2$.
6. If $y = tan^{-1}x$, show that $(1 + x^2)y_2 + 2xy_1 = 0$, hence prove that $(1 + x^2)y_{n+2} + 2(n + 1)xy_{n+1} + n(n + 1)y_n = 0$,